NISM ResearchAnalyst (XV)

Formula Sheet

EV = Value of common equity + value of non-controlling interest + Value of preferred capital + Debt – cash, cash equivalents, and financial investments.

Market cap = Price of a stock x Total number of outstanding shares

Earning Per Share = Net profit / Weighted average outstanding shares

Dividend Per Share = Total Dividends Paid / Shares Outstanding

P/S Ratio = Current Market Price (CMP) / Annual Net Sales per Share
Or
P/S Ratio = Market capitalization / Annual Net Sales

Coupon per period = Coupon rate X Face Value of the bond

Holding Period Return (HPR) = ((Income + (End of period value - Original value)) / Original value) * 100

Current yield (bond) = Annual coupon payment / Bond's current market value

Future Value (FV) = Present Value × (1 + r) ^ n

Where: PV = Present Value. r = Interest Rate (%) n = Number of Compounding Periods.
[^ raise to the power]

Present Value = FV / (1+i)^n

Working Capital = Current Assets - Current Liabilities

Profitability Ratios (Higher ratio, better firm)

EBITDA Margin = EBITDA / Net Sales

PAT Margin = PAT/ Net Sales

Return Ratios (Higher ratio, better firm)

Return on Equity (ROE) = PAT/ Net-worth

Net-worth = Equity Capital + Reserves & Surplus

Du Pont Analysis (ROE is further decomposed)

ROE = Net Profit Margin * Asset Turnover * Equity Multiplier
Or
PAT / Net-worth = (PAT / Net Sales) * (Net Sales / Fixed Assets) * (Fixed Assets / Net-worth)

$$ROE = \frac{Net\ profit}{Equity} = \frac{Net\ Profit}{Sales} * \frac{Sales}{Assets} * \frac{Assets}{Equity}$$

Return on Capital Employed (ROCE) = EBIT/ Capital Employed(Debt + Net-worth)

Leverage Ratios

D/ E Ratio = Long Term Debt / Net-worth **(Lower better)**

Interest Coverage Ratio = EBIT / Interest Expense **(Higher better)**

Liquidity Ratios (Compare ratio with Peers)

Current Ratio = Current Assets / Current Liabilities

Quick Ratio = (Current Assets – Inventories) / Current liabilities

Efficiency Ratios (Higher better)

Accounts Receivable Turnover = Revenue / Accounts Receivable

Accounts Payable Turnover = Purchases / Accounts Payable

Asset Turnover = Net Sales / Total Assets

Inventory Turnover = Sales / Inventory

Discounted Cash Flows Model

Dividend discount model

$$P = \frac{D1}{k - g}$$

D1 refers to dividend expected to be received at the end of the year

Free cash flow to equity model

FCFE = CFO – Fixed capital expenditure + Net borrowing

Value of the equity = Present value of FCFE during high growth phase +
Value of perpetual stream of FCFE after high growth phase
(referred to as terminal value)

Free cash flow to firm model (FCFF)

FCFF = CFO + Int(1 – Tax rate) – FCInv.

Value of the Firm = Present value of FCFF during high growth phase +
Value of perpetual stream of FCFF after high growth phase
(referred to as terminal value)

FCFF is discounted using the weighted average cost of capital (WACC).

Capital Asset Pricing Model (CAPM)

Ke (cost of equity) = Rf + β * (Rm – Rf)

Where:
Rf = Risk-Free Rate, (Rm – Rf) = Market risk premium (MRP) and β = Beta

Weighted Average Cost of Capital of the firm (WACC)

WACC = [Ke * Equity / (Equity+ Debt)] + [Kd * (1-Tax)* Debt / (Equity+ Debt)]

= [Ke * We] + [Kd * (1-Tx)*Wd]

Where:
Kd = Cost of Debt, Wd = Weight of Debt, Ke = Cost of Equity, We = Weight of Equity

Relative valuation

Earnings Based Valuation Metrics

Dividend Yield = Dividend per share (DPS) / Current price of the stock

Earning Yield = Earnings Per Share (EPS) / Current price of the stock

The reciprocal of Earning Yield is the popularly known Price to Earnings Ratio

PE Ratio = Market price per share / Earnings per share **(compare with peer group)**

Growth adjusted Price to Earnings Ratio = [Current Price of Stock / Earnings Per Share] / Growth rate

Or

PEG ratio = PE ratio/ Growth rate **(PEG ratio less than 1 is considered good)**

Where: Growth rate is the growth rate of earnings.

Enterprise Value to EBITDA Ratio = EV / EBITDA

Enterprise Value to EBIT Ratio = EV/ EBIT

Enterprise Value (EV) to Sales Ratio = EV / EBIT **(can never be negative)**

Assets-based Valuation Metrics

ROE

ROCE

Return on Invested Capital = Earnings / Invested Capital

Price to Book Value Ratio = Market capitalization / Balance sheet value of equity
Or
Price/Book ratio = Price per share / Book value per share

EV to Capital Employed ratio = Enterprise Value / Capital Employed (Total Equity + Total Debt)

Calculation of Simple, Annualized and Compounded Returns

Simple
RoI = (Total Returns/Total Cost) x 100

Where:
Total Cost = shares x price per share + commission fee,

Total Returns = Dividends + Sales Proceeds
Annualised
Absolute return is converted into annualized return by dividing it by the number of months/days that the investment was held and multiplying it by 12 months/365 days

Compounded Annual Growth Rate (CAGR)

{(End Value/Beginning Value)^(1/n)}-1

where n is the holding period in years.

Measuring risk

standard deviation

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Where $\bar{x}$ refers to the average rate of returns of the asset and n represents the number of observations in the sample.

Calculating risk-adjusted returns

Jensen's Alpha = Return on the portfolio – (Risk-free rate + β * market risk premium) **(Higher better)**
It measures as the excess return earned by a portfolio over and above the cost of equity that is calculated using CAPM

$$Sharpe\ ratio = \frac{Return\ on\ portfolio - Risk\ free\ rate}{Standard\ deviation}$$

Sharpe ratio measures the risk premium earned per unit of standard deviation
(Higher better)

$$Treynor\ ratio = \frac{Return\ on\ portfolio - Risk\ free\ rate}{Beta}$$

Treynor ratio measures the risk premium earned per unit of Beta.
(Higher better)

Loan to Value (LTV) = Loan Amount / Property Value